

CAPE MERCY, NU, Canada

WMO: 719750

Lat: 64.958N

Lon: 63.578W

Elev: 1911

StdP: 13.71

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.3	-25.1	N/A	N/A	N/A	N/A	N/A	N/A	58.6	1.1	52.9	-1.5	23.3	270	1.173

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	55.4	N/A	52.3	N/A	49.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 48.2	41.8	36.4	DB	-30.8	58.7	4.9	3.1	-34.3	61.0	-37.1	62.8	-39.9	64.5	-43.4	66.8	
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	-6.3	-9.2	-3.6	7.2	20.7	31.9	40.7	39.4	29.8	21.7	12.4	1.6
	DBStd	19.00	11.15	11.10	11.47	9.67	7.16	5.95	6.74	5.60	4.70	5.99	7.85	11.12
	HDD50	12546	1746	1658	1662	1284	908	544	301	332	606	876	1128	1501
	HDD65	18005	2211	2078	2127	1734	1373	993	755	793	1056	1341	1578	1966
	CDD50	16	0	0	0	0	0	0	12	4	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	15.3	16.1	15.8	15.6	15.1	14.5	14.7	13.5	13.8	15.6	17.3	15.6	15.9
	PrecAvg	23.7	1.4	0.9	0.9	1.1	1.3	1.7	2.3	3.1	3.2	3.2	2.5	2.1
(15) Precipitation	PrecMax	31.7	3.3	3.6	2.4	2.3	2.6	3.4	5.2	4.8	5.2	5.1	6.4	4.7
	PrecMin	13.1	0.4	0.3	0.2	0.2	0.4	0.8	0.8	1.3	0.8	1.8	0.9	0.6
	PrecStd	4.6	0.8	0.7	0.6	0.5	0.5	0.6	0.9	0.9	1.0	0.7	1.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	23.8	27.8	31.7	44.3	53.2	61.5	57.8	47.1	46.9	34.6	29.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	21.0	18.3	22.4	27.6	38.0	47.5	57.2	53.8	42.2	36.3	28.1	24.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	15.3	11.8	17.6	24.3	32.7	44.2	54.2	50.9	38.6	31.9	25.2	20.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	10.4	6.8	13.0	20.4	29.3	41.1	51.0	48.2	36.0	29.3	22.6	16.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% DB	MDBR	8.8	8.4	8.8	8.2	7.0	6.3	7.8	7.3	5.0	5.5	6.7	7.9
		MCDBR	13.0	12.6	12.1	9.5	10.8	10.0	10.3	10.3	9.5	10.0	10.7	11.4
(35) Mean Daily Temperature Range	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.142	0.201	0.231	0.276	0.296	0.302	0.315	0.306	0.280	0.246	0.334	0.142
	taud		1.877	2.165	2.205	2.108	2.126	2.257	2.439	2.532	2.541	2.375	3.014	1.904
	Ebn at Noon		138	237	277	284	288	289	282	276	261	215	134	49
	Edh at Noon		7	17	26	37	41	37	30	25	20	14	6	2
(44) All-Sky Solar Radiation	RadAvg		45	288	846	1561	2042	2042	1747	1169	722	293	62	12
	RadStd		3	14	36	61	95	167	169	119	73	26	5	1

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+3.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	+3.44	+3.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page